



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

GENBE302

BASIC ELECTRONICS

APPLY BASIC ELECTRONICS

Competence

RQF Level: 3

Learning Hours



Credits: 6

Sector: ICT & MULTIMEDIA

Trade: COMPUTER SYSTEM AND ARCHITECTURE

Module Type: General

Curriculum: ICTNIT3001 -TVET Certificate III in Computer System and Architecture

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Purpose statement	This general module describes the skills, knowledge and attitude required to use basic electronics components. This module is intended to prepare student pursuing TVET Level 3 in Computer System and Architecture. Upon completion of this module, the trainees will be able to describe, apply and test passive and active components. In addition, the trainees will be able to identify, use different equipment used in electronic workshop.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	100%
	Practical work:		70%		70%	
	● Group project and presentation	20%				
	● Individual project /Work	50%				

Elements Performance Criteria	
Elements of competence	Performance criteria
1. Apply passive components	1.1. Passive components are clearly described according to their types and features
	1.2. Passive components are adequately selected according to the work to be done
	1.2. Passive components circuits are properly tested according to their types and applications
2. Apply Active components	2.1. Active components are clearly described according to their types and features
	2.2. Active components are properly selected according to the work to be done
	2.3. Integrated Circuits (IC) are properly applied according to their types and applications
	2.4. Active components circuits are properly tested based on their types and allocation
3. Use electronic workshop equipment	3.1. Electronic workshop equipment's are identified according to their types and functions
	3.2. Electronic workshop equipment's are adequately used according to their functions
	3.3. Electronic workshop equipment's are properly tested according to their features

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Apply passive components 2. Apply active components 3. Use electronic workshop equipment
Learning outcome 1: Apply passive components	Learning hours: 20
Indicative content	
● Introduction to passive components ● Description of resistors ✓ Resistor specification ✓ Resistivity ✓ Ohms law ✓ Types of resistors and their applications ● Fixed resistor ● Variable resistors ✓ Resistor Colour code ● Application of resistor ✓ Circuit connection of resistor ✓ Voltage divider rule ✓ Measuring resistance ✓ Resistor testing ● Description of capacitors ✓ capacitor specification ✓ The types of capacitors and their applications ✓ capacitance colour coding ● application of capacitor ✓ Soldering process ✓ Circuit connection of capacitor ✓ Measuring capacitance ✓ Capacitor testing ● Description of inductors ✓ The types of inductors and their applications	

- ✓ Inductor specification
- ✓ Mutual inductance theorem

- **Application of inductor**

- ✓ Circuit connection of inductor
- ✓ Measuring inductance
- ✓ Inductor testing

Resources required for the learning outcome

Equipment	Multimeter, LCR-meter, LC-meter ,computer, projector, PPE,
Materials	Resistors, capacitor , inductor , jumper wires, soldering tin
Tools	Breadboard or PCB, soldering iron, disordering pump
Facilitation techniques	● Demonstration ● Group discussion ● Individual and group work ● Practical exercises
Formative assessment methods	● Written assessment ● Oral presentation ● Product assessment

Learning outcome 2: Apply Active components

Learning hours: 25

Indicative content

- **Introduction to semiconductor devices**

- **Description of diodes**

- ✓ Working principle of diodes
- ✓ The types of diodes
- ✓ Biasing a diode
- ✓ VI characteristics of diode

- **Applications of diodes**

- ✓ Rectify
- ✓ Clipping
- ✓ Clamper
- ✓ Voltage regulator
- ✓ Diode testing

- **Description of transistors**

- ✓ Transistor structure
- ✓ Working principle of transistors
- ✓ The types of transistors
 - BJT
 - FET
- ✓ Operating regions
- ✓ Bipolar junction Transistor configuration
- ✓ Characteristics of transistor
- ✓ Transistor & diode coding system

- **Applications of transistors**

- ✓ Switching
- ✓ Amplifying
- ✓ Transistor testing

- **Description of integrated circuits (IC)**

- ✓ Integrated circuit
- ✓ Chipset
- ✓ The types of integrated circuits
- ✓ Pin configuration
- ✓ Packaging of integrated circuits
- ✓ Marking/codification of an IC

- **Applications of integrated circuits**

- ✓ Soldering procedure
- ✓ Timing
- ✓ Counter
- ✓ Amplification
- ✓ Regulator
- ✓ IC testing

Resources required for the indicative content		
Equipment	Multimeter, RCL meter, LC meter, oscilloscope, power supply, function generator, Computer, projector, soldering station ,IC tester,	
Materials	Diodes, transistors, integrated circuits(555,4017,741,317) , jumper wires , soldering tin ,electrical power	
Tools	Breadboard, soldering iron, disordering pump, toolbox set ,lab manual	
Facilitation techniques	• Individual and group work • Group discussion • Demonstration • Practical exercises	
Formative assessment methods	• Written assessment • Oral presentation • Product based assessment	
Learning outcome 3: Use electronics workshop equipment		Learning hours: 15
Indicative content		
• Description of different electronics workshop equipment ✓ Identification of electronic workshop equipment ✓ Types of equipment based on their function ✓ Identification of main parts of equipment and their function ✓ Technical specification of equipment based on their function • Use of electronics workshop equipment ✓ Resistance, voltage and current measurement ✓ Capacitance & inductance measurement ✓ Frequency & amplitude measurement ✓ Testing electronic components ✓ Engineering • Testing electronics workshop equipment		

Resources required for the indicative content	
Equipment	Multimeter, RLC meter, LC meter, oscilloscope, function generator ,power supply ,Computer, projector, soldering station,
Materials	Electronic components(passive &active compnents), jumper wires, breadboard, soldering wires, soldering tin , fuse
Tools	Soldering iron, disordering pump pliers, soldering stand ,screwdrivers, magnifying glass
Facilitation techniques	• Demonstration • Individual and group work • Practical exercise • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral presentation • Performance assessment • Product assessment

References:

1. Buchla, D. (1998). *Experiments in Electronics Fundamentals and Electrical circuits Fundamentals* (Vol. 4th). Prentice Hall
2. MALVINO, A., & BATES, D. (2016). *ELECTRONIC PRINCIPLES* (Vol. 8th). New York,: McGraw-Hill.
3. Principles and Applications of Electrical Engineering – Chapter 9, pp. 495 – 620. 5th Edition. Rizzoni, Giorgio. McGraw-Hill, 2007.
4. Type 5MC Metallized Polycarbonate Capacitor, Electronic Concepts, Inc., Box 1278, Eatontown, NJ, 07724, (908) 542-7880.